SIMATS ENGINEERING

(SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – EXPERIMENTATION

Course Code:	ITA03	**Course Title:**	Mobile Computing
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Experimentation
Weightage:	30 % of CIA	**Total Marks:**	100 Marks
CO2 Statement:	Compare mobile communication technologies, including multiple access techniques and network evolution, based on performance and applications.		
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Section / Batch:	BE CSE/2023	**Date:**	26-10-2026

Code of Conduct

I A . M a n o j R e d d y (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: A.Manoj Reddy

Instructions

- This test contains FIVE questions.
- Duration: 60 minutes (homework).

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

ANNEXURE A – QUESTIONS AND SOLUTIONS

Each solution follows the evaluation rubric: A. Design Quality • B. Technical Analysis • C. Correctness of Concepts • D. GitHub Repository & Documentation • E. Creativity and Visualization

Question 1: Figma-Based Mobile IP Communication Workflow Analysis (BL4 – Analyze)

Problem Statement

Design an interactive Figma prototype illustrating the complete workflow of Mobile IP communication when a Mobile Node moves from its Home Network to a Foreign Network, including the HA, FA, CN, CoA, registration process and packet tunneling, and analyse the communication process.

A. Design Quality (Figma Design) *[20 Marks]*

The design is built as four connected Figma frames; the frames are linked with prototype arrows so they can be played in sequence.

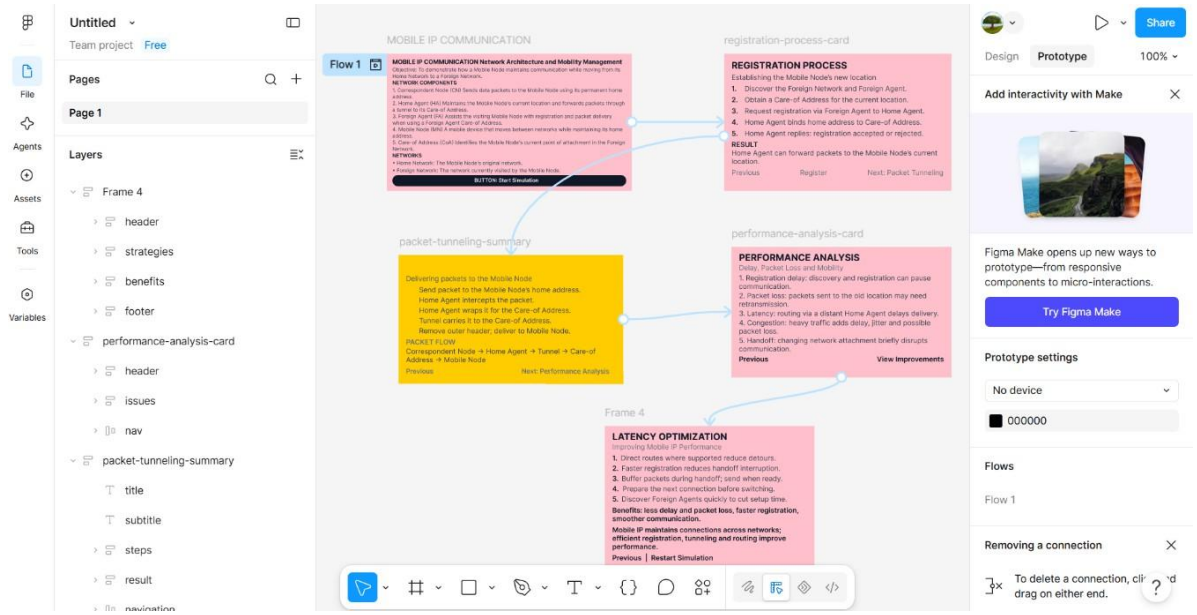


Figure 1: *Figma design frames – Mobile IP workflow (Frame 1 to Frame 4)*

Complete design board

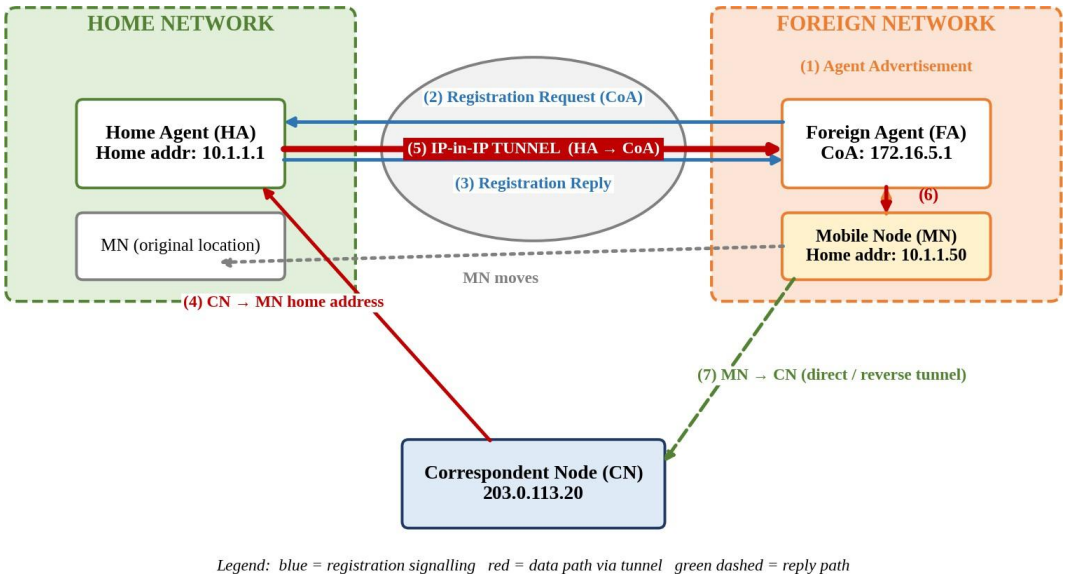


Figure 2: *Figma design output – Mobile IP workflow (registration and packet tunneling)*

Design description

- **Layout:** One flow board with the Home Network (green) on the left, the Foreign Network (orange) on the right, and the Internet and CN in the centre.
- **Flow:** Numbered steps (1)–(7) show agent advertisement, registration request and reply, interception, tunneling, delivery and return traffic.
- **Colour coding:** Blue = registration signalling, red = tunneled data path, green dashed = return path, grey dotted = node movement.
- **Prototype:** Frames per step, linked with Figma prototype arrows (On click → Navigate to) so the sequence can be played from the first to the last step.

Workflow summary

- **Agent discovery:** FA advertises; the MN learns it is in a foreign network and obtains a CoA.
- **Registration:** MN → FA → HA request binds the home address to the CoA; the HA replies.
- **Interception and tunneling:** HA intercepts packets for the home address and encapsulates them to the CoA.
- **Delivery:** FA decapsulates and delivers to the MN; the MN replies directly or through a reverse tunnel.

B. Technical Analysis *[30 Marks]*

(a) Purpose of each network component

Component	Purpose
Mobile Node (MN)	Device that changes its point of attachment while keeping a permanent home IP address.
Home Agent (HA)	Home-network router that keeps the binding table, intercepts packets for the MN and tunnels them to the CoA.
Foreign Agent (FA)	Visited-network router that advertises, relays registration, decapsulates tunneled packets and delivers them to the MN.
Correspondent Node (CN)	Peer host that addresses packets only to the MN's home address.
Care-of Address (CoA)	Temporary address identifying the MN's current location.
Registration Process	Signalling that creates or refreshes the home-address-to-CoA binding.
Packet Tunneling	Encapsulation that carries packets from the HA to the CoA without altering the original datagram.

(b) Points where packet delay or packet loss may occur

Point	Cause	Effect
Movement / handoff	MN detaches from the old link before registration completes.	Packets sent to the old location are dropped.
Registration round trip	Request and reply cross the Internet to the HA and back.	Handoff latency grows with HA distance.
HA processing	Binding lookup and encapsulation for every packet.	Queuing delay; HA becomes a bottleneck.
Triangular routing	CN → HA → FA → MN is longer than the direct path.	Higher delay and extra bandwidth.
Tunnel overhead	Extra 20-byte outer IP header per packet.	Lower payload efficiency; possible fragmentation.
Binding expiry	Registration lifetime ends or a reply is lost.	Traffic lost until re-registration.

(c) Effect of node mobility on communication performance

- Frequent movement causes repeated registrations, raising signalling overhead and packets lost per handoff.
- TCP throughput falls because losses are read as congestion and the window shrinks.
- Latency rises with distance from the HA because every packet takes the home-network detour.
- At high speed the registration time can exceed the time spent in a cell, making the connection unstable.

(d) Possible improvements to reduce latency

- **Route optimisation –** CN caches the binding and sends directly to the CoA, removing triangular routing.
- **Hierarchical Mobile IP –** A local mobility anchor handles intra-domain moves so the distant HA is not contacted.
- **Fast handover –** Pre-register with the next FA and buffer packets at the old FA during the switch.
- **Co-located CoA / nearer HA –** Use a co-located CoA and regional HAs closer to users.

C. Correctness of Mobile Networking Concepts *[20 Marks]*

Concept	Correct technical statement
Mobile IP	IETF standard (RFC 5944 for IPv4, RFC 6275 for IPv6) that lets a node keep one home address while changing network.
Agent discovery	Uses ICMP Router Discovery extended with mobility agent advertisements.
Registration	Request and reply messages are sent over UDP (port 434) with authentication extensions.
Tunneling	IP-in-IP encapsulation (RFC 2003); the FA or MN removes the outer header.
CoA types	FA CoA (FA address) or co-located CoA (address obtained by the MN, e.g. via DHCP).
Triangular routing	Forward path goes through the HA while the return path is direct; it is solved by route optimisation or reverse tunneling.

D. GitHub Repository Organization & Documentation *[15 Marks]*

Repository structure

Folder / File	Contents
/design/	mobile-ip-workflow.fig (editable Figma file)
/exports/	PNG and PDF exports of every frame and the full prototype
/docs/	analysis-report.pdf (components, delay points, improvements)
/assets/	icons, colour palette, fonts
README.md	Project overview, how to open the prototype, screenshots

README.md contents

- Title, objective and problem statement.
- Link to the Figma prototype and instructions to play it.
- Exported workflow image and component description.
- Analysis summary and references (RFC 5944, RFC 6275).

E. Creativity and Visualization *[15 Marks]*

- Colour-coded network zones and message types for quick reading.
- Animated packet movement between frames using Figma Smart Animate.
- Clickable component cards showing the purpose of the HA, FA, CN and CoA.
- Step numbers and a legend so the workflow can be followed without extra explanation.

Question 2: Comparative Network Architecture Design Using Figma (BL4 – Analyze)

Problem Statement

Create a comparative architectural diagram showing a Cellular Wireless Network and a Mobile Ad Hoc Network (MANET) operating in the same smart city environment, and analyse both architectures.

A. Design Quality (Figma Design) *[20 Marks]*

The design is built as four connected Figma frames; the frames are linked with prototype arrows so they can be played in sequence.

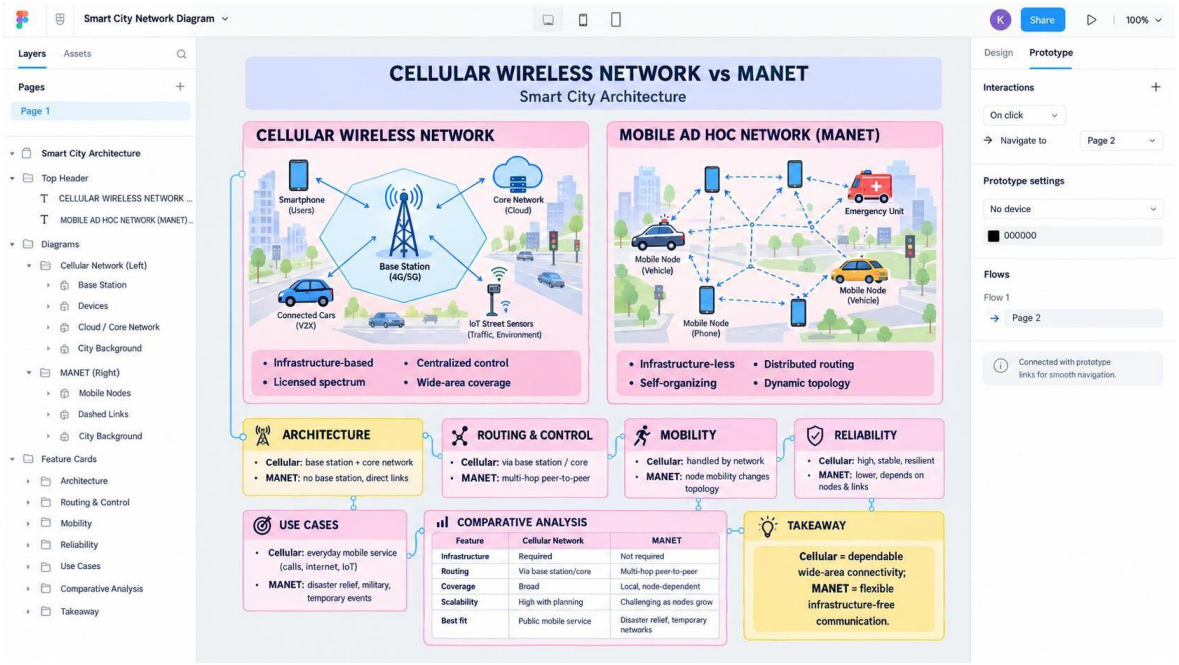


Figure 3: *Figma design frames – Cellular Network and MANET comparison (Frame 1 to Frame 4)*

Complete design board

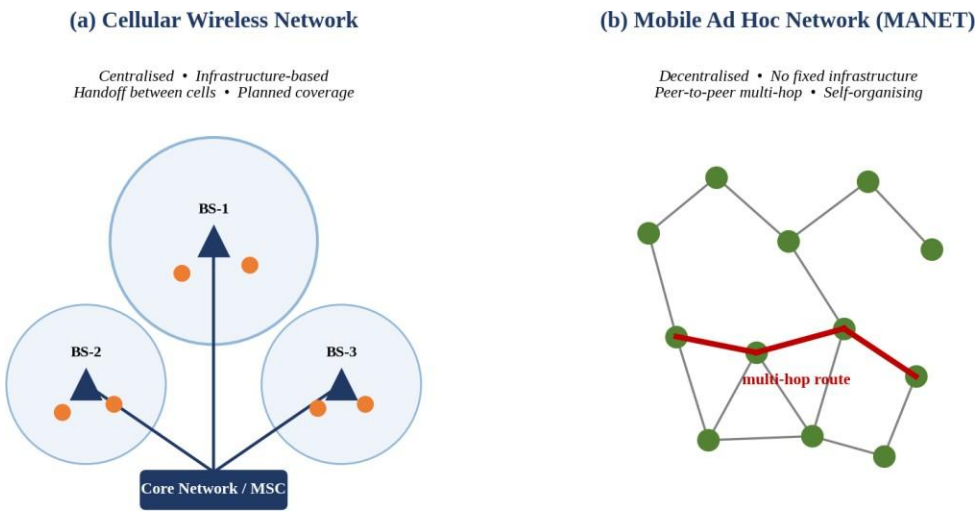


Figure 4: *Figma design output – Cellular Network and MANET in a smart city*

Design description

- **Layout:** Two panels with matching scale: cells with base stations and a core network on the left, peer nodes with a multi-hop route on the right.
- **Annotations:** Annotation callouts under each panel list the advantages and limitations.
- **Colour coding:** Blue for cellular elements, green for ad hoc nodes, red for the active route.

B. Technical Analysis *[30 Marks]*

Parameter	Cellular Wireless Network	Mobile Ad Hoc Network (MANET)
Infrastructure dependency	Fully dependent on base stations, backhaul and core network.	No fixed infrastructure; nodes act as hosts and routers.
Routing mechanism	Centralised routing and mobility management through the core network.	Distributed routing by AODV, DSR, DSDV or OLSR.
Scalability	Scales by adding cells and reusing spectrum.	Degrades as node count grows due to control overhead and contention.
Fault tolerance	A base-station or core failure affects a cell; mitigated by redundancy.	No single point of failure; routes are rebuilt around failed nodes.
Mobility support	Seamless handoff between cells at vehicular speeds.	Supported, but topology change causes route breaks.
Suitable applications	Voice, broadband, IoT metering, city-wide coverage.	Disaster recovery, V2V messaging, temporary events, military.

Advantages and limitations

	Advantages	Limitations
Cellular	Planned coverage, high reliability and QoS, managed security, seamless handoff.	High deployment and spectrum cost, infrastructure dependence, congestion in crowded areas.
MANET	Rapid deployment, low cost, self-healing, works without infrastructure.	Limited bandwidth and battery, security threats, poor scalability, unstable routes.

Conclusion: In a smart city, cellular networks should carry the backbone services while MANETs complement them for emergency response, vehicle-to-vehicle messaging and areas without coverage.

C. Correctness of Mobile Networking Concepts *[20 Marks]*

Concept	Correct technical statement
Cellular concept	Coverage is divided into cells; frequency reuse increases capacity.
Handoff	Call or session is transferred between cells without interruption (hard or soft handoff).

Concept	Correct technical statement
Mobility management	Location databases (HLR/VLR in GSM, MME/HSS in LTE) track the user's current cell.
MANET	Self-configuring network of mobile nodes with multi-hop wireless links and no central controller.
Medium access	MANETs commonly use IEEE 802.11 CSMA/CA, which suffers from hidden and exposed terminal problems.

D. GitHub Repository Organization & Documentation *[15 Marks]*

Repository structure

Folder / File	Contents
/design/	cellular-vs-manet.fig / .drawio (editable files)
/screenshots/	Exported PNG of both architectures and annotations
/report/	comparison-report.pdf
/assets/	icons and shapes used
README.md	Overview, comparison summary, usage

README.md contents

- Objective and scenario description.
- Screenshots with the comparison table.
- Advantages and limitations summary.
- Tools used and how to open the design files.

E. Creativity and Visualization *[15 Marks]*

- Same smart-city scenario shown in both architectures for a fair side-by-side comparison.
- Icons for base stations, vehicles and sensors, with the multi-hop route highlighted.
- Callout annotations placed directly on the diagram instead of in a separate legend.

Question 3: Routing Protocol Visualization and Analysis (BL4 – Analyze)

Problem Statement

Develop a visualisation of one Proactive routing protocol (DSDV) and one Reactive routing protocol (AODV) in a mobile wireless network and analyse route discovery, maintenance, overhead, delay and performance under high mobility.

A. Design Quality (Figma / Visualization) *[20 Marks]*

The design is built as four connected Figma frames; the frames are linked with prototype arrows so they can be played in sequence.

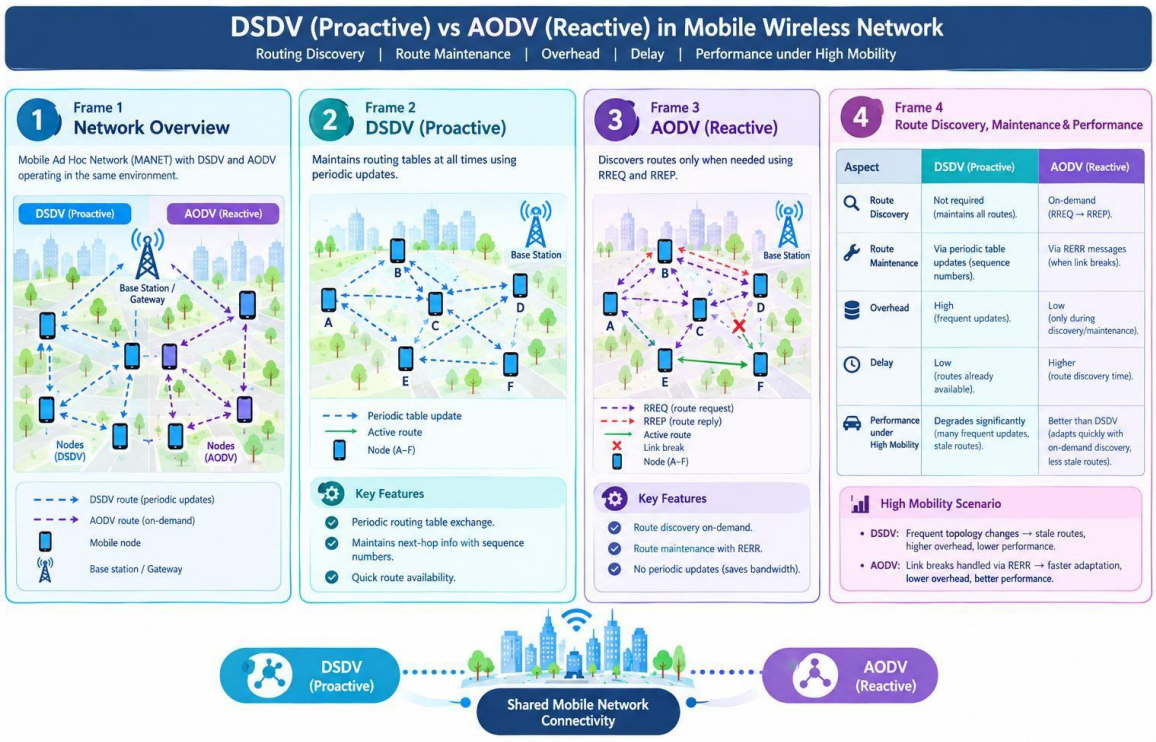


Figure 5: *Figma design frames – DSDV and AODV routing visualization (Frame 1 to Frame 4)*

Complete design board

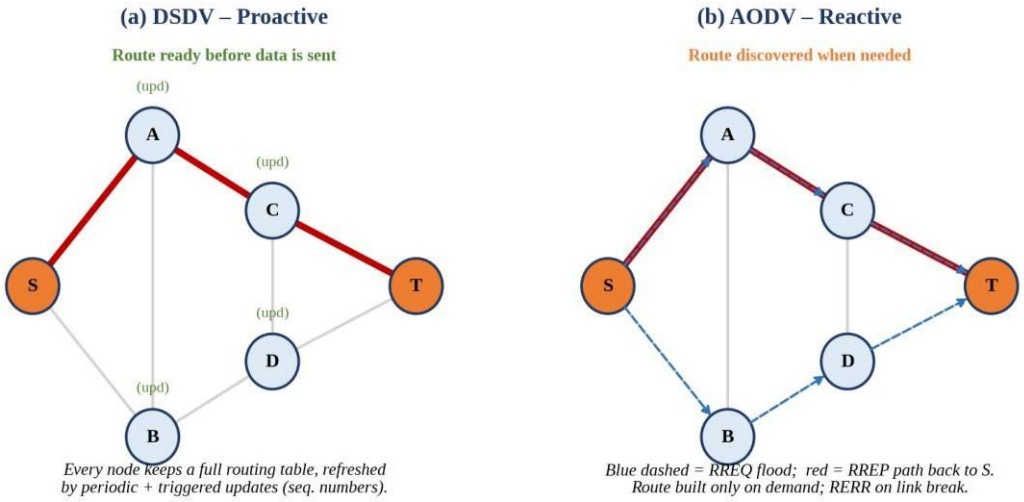


Figure 6: *Figma design output – DSDV (proactive) and AODV (reactive) on the same topology*

Design description

- **Layout:** Same six-node topology used for both protocols so that behaviour is directly comparable.
- **Colour coding:** Red = selected route, blue dashed = RREQ flood, green markers = periodic table updates.
- **Interaction:** Frames step through route discovery, data transfer and a link break to show maintenance.

B. Technical Analysis *[30 Marks]*

DSDV: every node keeps a table of next hop, metric and sequence number for all destinations, exchanged periodically and on change, so routes always exist.

AODV: a source broadcasts a Route Request (RREQ); the destination or a node with a fresh route replies with a Route Reply (RREP) along the reverse path; link breaks are reported with Route Error (RERR).

Aspect	DSDV (Proactive)	AODV (Reactive)
Route discovery	None needed; routes pre-computed.	RREQ flood and unicast RREP when needed.
Route maintenance	Periodic and triggered updates with sequence numbers.	Hello messages and RERR; re-discovery on demand.
Routing overhead	High and constant even without traffic; grows with size.	Low when traffic is light; bursts of flooding.
Delay to establish communication	Minimal, route exists.	One route-discovery round trip.
Performance under high mobility	Tables go stale; heavy updates and loss.	Better, as only active routes are repaired.

Comparative summary by mobility condition

Mobility condition	Better protocol	Reason
Low mobility, dense traffic	DSDV	Stable topology keeps update overhead modest; no start-up delay.
Moderate mobility	AODV	Overhead is spent only on routes in use.
High mobility	AODV	DSDV updates cannot keep pace; AODV repairs only active paths.
Delay-critical, small network	DSDV	Routes are always ready.

C. Correctness of Mobile Networking Concepts *[20 Marks]*

Concept	Correct technical statement
DSDV	Table-driven protocol based on the Bellman–Ford distance-vector algorithm; destination sequence numbers prevent routing loops.
AODV	On-demand distance-vector protocol defined in RFC 3561; uses RREQ, RREP, RERR and destination sequence numbers.
Proactive vs reactive	Proactive protocols maintain routes continuously; reactive protocols build routes only when data is to be sent.
Expanding ring search	AODV limits RREQ flooding by gradually increasing the TTL.

D. GitHub Repository Organization & Documentation *[15 Marks]*

Repository structure

Folder / File	Contents
/src/	visualisation source code or Figma/Mermaid files
/screenshots/	DSDV and AODV step-by-step images
/analysis/	analysis-report.pdf with comparison
README.md	How to run or open the visualisation

README.md contents

- Objective and protocols covered.
- Instructions to run the visualisation.
- Screenshots of each stage.
- Analysis summary and references (RFC 3561, Perkins & Bhagwat 1994).

E. Creativity and Visualization *[15 Marks]*

- Side-by-side panels on one topology, with a step slider for the protocol stages.
- Animated RREQ flooding and a highlighted RREP return path.
- A simulated link failure showing how each protocol reacts.

Question 4: Mobile IP Application Design for Smart Healthcare (BL4 – Analyze)

Problem Statement

Design a Smart Healthcare Mobile IP system in Figma that keeps doctors, ambulances, hospitals and patients connected while moving across different wireless networks, and analyse its operation.

A. Design Quality (Figma Design) *[20 Marks]*

The design is built as four connected Figma frames; the frames are linked with prototype arrows so they can be played in sequence.

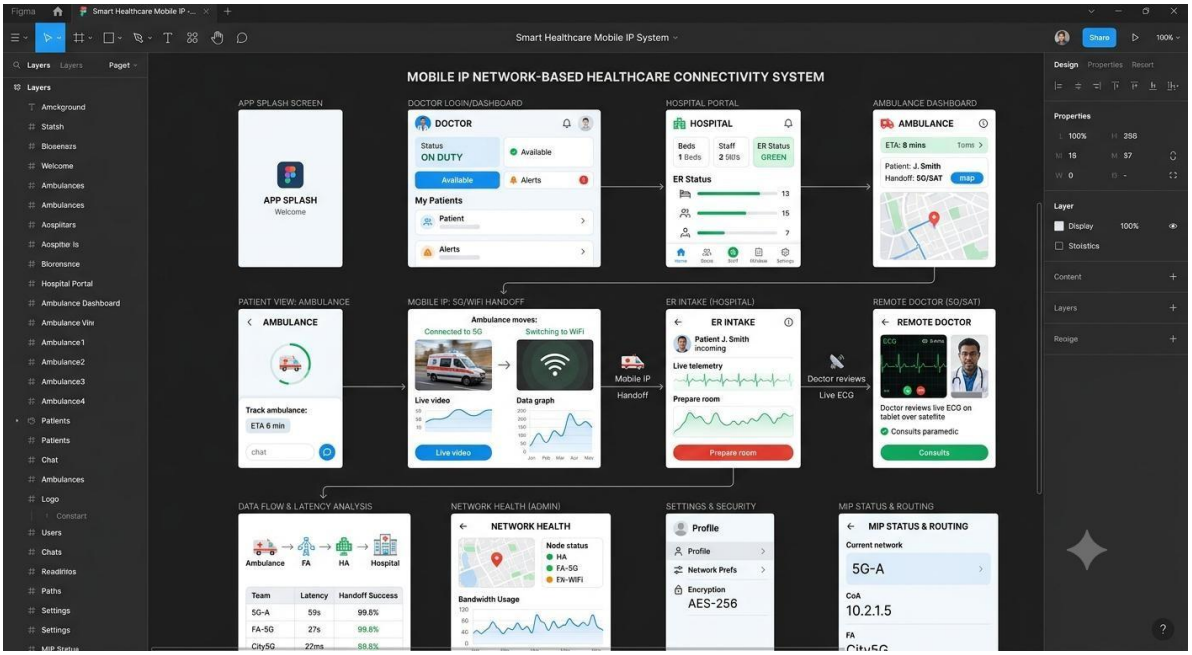


Figure 7: *Figma design frames – Smart Healthcare mobile app screens (Frame 1 to Frame 4)*

Complete design board

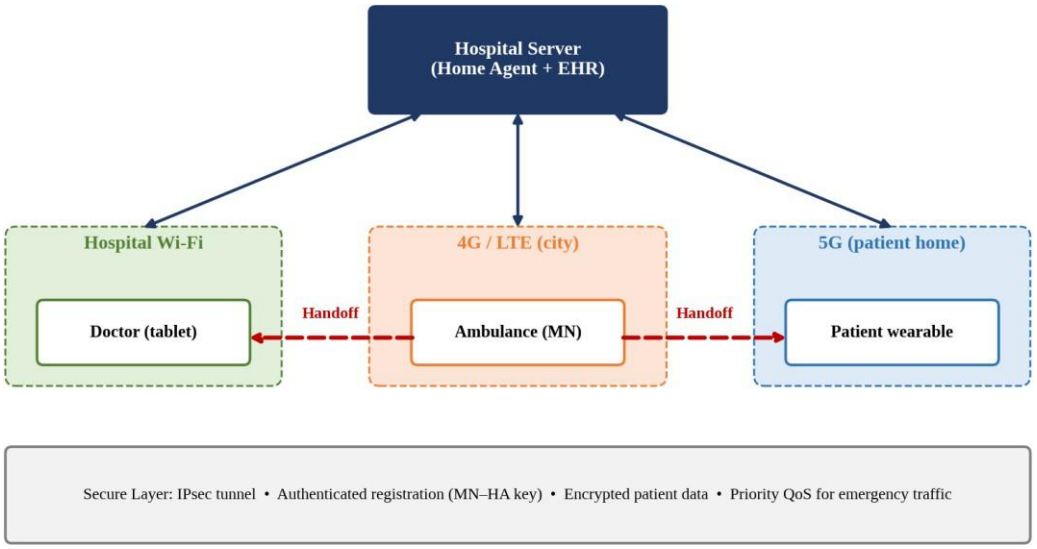


Figure 8: *Figma design output – Smart Healthcare Mobile IP architecture with handoff*

Prototype screen navigation

Screen	Name	Mobility scenario demonstrated
1	Login & Authentication	Mutual authentication with the hospital Home Agent.
2	Dashboard	Live network status, current CoA and connection quality.
3	Ambulance Tracking	Ambulance moves from 4G to hospital Wi-Fi; vitals keep streaming.
4	Doctor Consultation	Doctor roams between wards with a live video session.
5	Patient Monitoring	Wearable moves from home Wi-Fi to 5G, reachable at its home address.
6	Handoff & Alerts	Handoff events, packet loss and security warnings.

B. Technical Analysis *[30 Marks]*

(a) How Mobile IP maintains uninterrupted communication

Each device keeps a permanent home address. When it moves, it registers a new care-of address with the Home Agent (hospital server), which tunnels packets to the current location, so sessions continue across Wi-Fi, 4G and 5G.

(b) Challenges during handoff between networks

- Handoff delay causes brief loss of vital-sign packets and interrupts live video.
- Different bandwidth, latency and IP versions affect QoS.
- Triangular routing adds delay when the ambulance is far from the Home Agent.
- Frequent re-registration drains the wearable's battery.

(c) Security vulnerabilities during mobility

Vulnerability	Risk	Countermeasure
Spoofed registration	Attacker redirects patient traffic to a false CoA.	Authenticated registration with MN–HA key.
Replay attack	Old registrations reused to hijack a session.	Timestamps or nonces.
Eavesdropping on tunnel	Patient data exposed on public networks.	IPsec / TLS encryption.
Rogue Foreign Agent	Fake FA captures or alters packets.	FA authentication and certificates.
DoS on Home Agent	Flooded registrations exhaust the HA.	Rate limiting and redundant HAs.

(d) Design improvements to increase reliability

- Redundant Home Agents with failover and regional agents.
- Fast handover with buffering, and multi-homing (Wi-Fi and cellular together) for the ambulance.
- Route optimisation and priority QoS marking for emergency traffic.
- Local mobility anchors to reduce registration frequency.

C. Correctness of Mobile Networking Concepts *[20 Marks]*

Concept	Correct technical statement
Mobile IP in healthcare	Permanent home address for each device gives a stable identity for records and monitoring systems.
Vertical handoff	Handoff between different technologies (Wi-Fi ↔ 4G ↔ 5G), more complex than horizontal handoff.
Mobile IP security	Registration is authenticated; data confidentiality is provided by IPsec (RFC 4301).
Route optimisation (MIPv6)	Return routability procedure lets the CN send directly to the CoA, avoiding the HA detour.
Patient data protection	Health data must be encrypted in transit and access-controlled according to applicable privacy regulations.

D. GitHub Repository Organization & Documentation *[15 Marks]*

Repository structure

Folder / File	Contents
/design/	smart-healthcare.fig (editable prototype)
/exports/	PDF and PNG of all screens, exported prototype
/docs/	analysis and design documentation

Folder / File	Contents
README.md	Scenarios, navigation guide, screenshots

README.md contents

- Objective and users (doctor, ambulance, hospital, patient).
- Prototype link and navigation guide.
- Mobility scenarios and screenshots.
- Security and reliability analysis summary.

E. Creativity and Visualization *[15 Marks]*

- Real-world scenario flow: ambulance trip → hospital arrival → doctor consultation.
- Live network indicator showing the current network and CoA during handoff.
- Red emergency alerts and a clear colour system for each network type.

Question 5: Wireless Network Decision Dashboard Using GitHub Project (BL4 – Analyze)

Problem Statement

Create a dashboard that helps network engineers decide whether a scenario should use Mobile IP, Cellular Networks or Ad Hoc Networks, analysing multiple parameters and justifying the recommendation.

A. Design Quality (Figma / Programming Tool) *[20 Marks]*

The design is built as four connected Figma frames; the frames are linked with prototype arrows so they can be played in sequence.

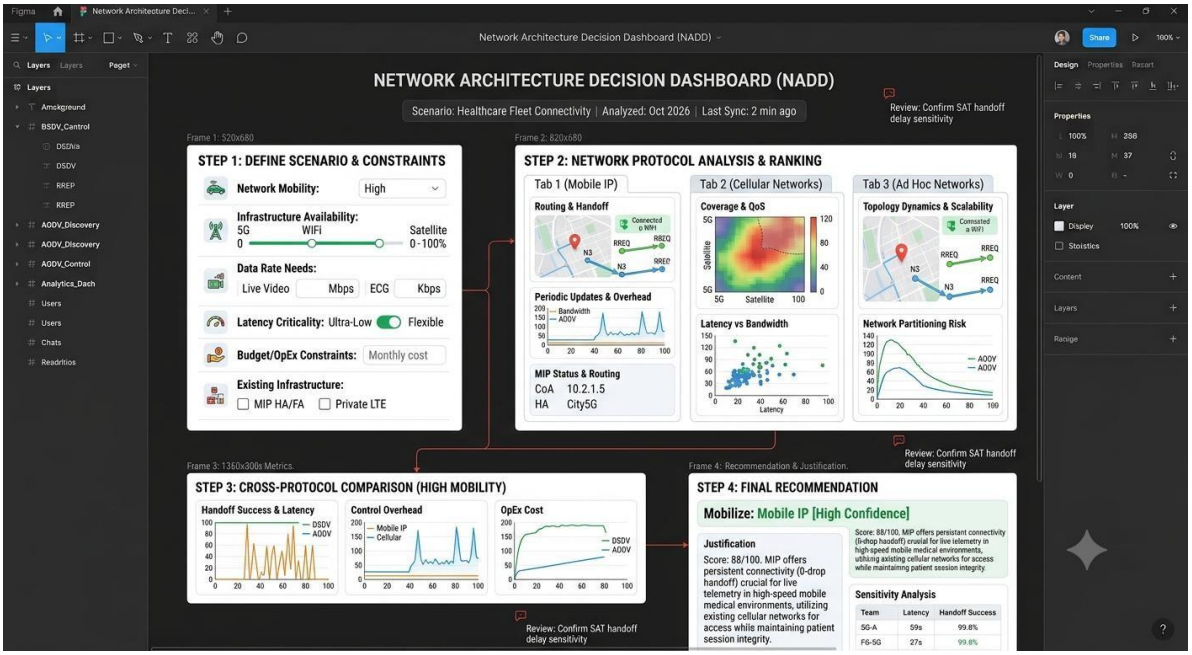


Figure 9: *Figma design frames – Wireless Network Decision Dashboard (Frame 1 to Frame 4)*

Complete design board

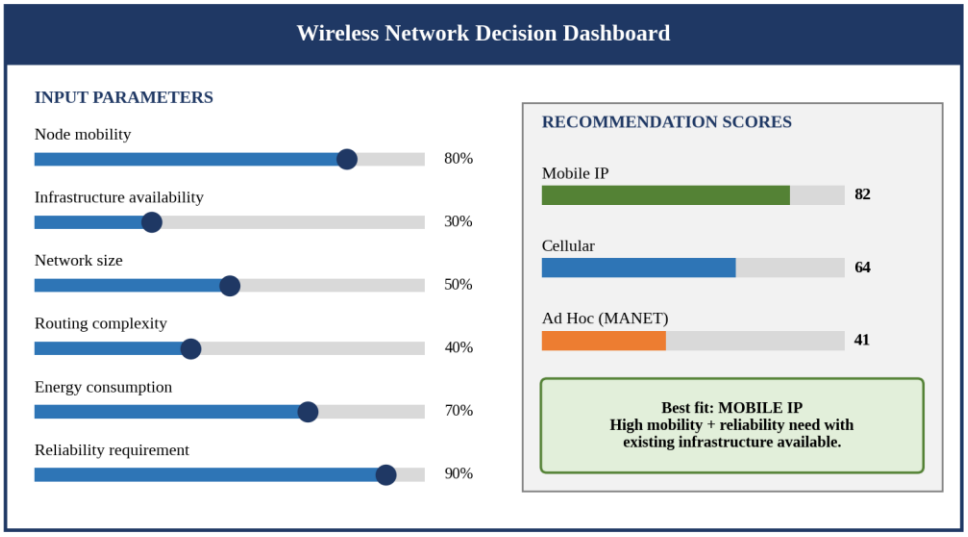


Figure 10: *Figma design output – Wireless Network Decision Dashboard*

Design description

- **Layout:** Inputs on the left (six sliders), results on the right (scores and recommendation card).
- **Colour coding:** Navy header, blue controls, green recommendation highlight for clear hierarchy.

B. Technical Analysis *[30 Marks]*

Parameter	Favours Mobile IP	Favours Cellular	Favours Ad Hoc
Node mobility	Medium–high, across IP subnets	Medium–high, wide area	High, local area
Infrastructure availability	IP infrastructure available	Operator coverage available	None or damaged
Network size	Small to large	Large, city scale	Small to medium
Routing complexity	Low (handled by agents)	Low (handled by network)	High (distributed)
Energy consumption	Moderate	Low on device, high on network	High (nodes forward traffic)
Reliability requirement	High, session continuity	High, managed QoS	Moderate, best effort

Recommendation logic

Each parameter is normalised to 0–100 and multiplied by a weight for each technology. The weighted sum gives each score, the highest score is recommended, and the justification lists the parameters that contributed most.

Sample scenario

Inputs: mobility 80 %, infrastructure 30 %, size 50 %, routing complexity 40 %, energy 70 %, reliability 90 %. Result: Mobile IP scores highest because high mobility and a high reliability requirement are met by a permanent address and session continuity. Cellular is a close alternative if coverage improves; Ad Hoc ranks lowest because of the reliability requirement and the energy cost of multi-hop forwarding.

C. Correctness of Mobile Networking Concepts *[20 Marks]*

Concept	Correct technical statement
Mobile IP	Network-layer mobility with a fixed home address; needs IP infrastructure and agents.
Cellular network	Operator-managed, infrastructure-based network with handoff and QoS.
Ad hoc network	Infrastructure-less, multi-hop network where every node routes traffic.

Concept	Correct technical statement
Weighted scoring	A multi-criteria decision method where each parameter's contribution is scaled by its weight.

D. GitHub Repository Organization & Documentation *[15 Marks]*

Repository structure

Folder / File	Contents
/src/ or /design/	dashboard source code (HTML/CSS/JS or React) or Figma project
/screenshots/	dashboard views for sample scenarios
/docs/	analysis-report.pdf, scoring rules
README.md	Setup, usage and scoring explanation

README.md contents

- Objective and features.
- Setup and run instructions.
- Parameter definitions and scoring logic.
- Screenshots and sample results.

E. Creativity and Visualization *[15 Marks]*

- Live-updating score bars as the sliders move.
- Plain-language justification generated with every recommendation.
- Colour-coded recommendation card and preset scenarios (disaster zone, campus, city).